

Israt Tamanna Niharika/ B.Sc.

University of Rajshahi, RAJSHAHI, BANGLADESH.

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EDUCATION

Program	Institution	CGPA	Year
MS- Genetic Engineering and Biotechnology	University of Rajshahi		05/08/2026 - 11/08/2027
B.Sc.- Genetic Engineering and Biotechnology	University of Rajshahi	3.66/4.00	12/21/2021 - 04/20/2026
HSC (Higher Secondary Certificate)	Cox's Bazar Government College	5.00/5.00	2020
SSC (Secondary School Certificate)	Cox's Bazar Govt. Girls' High School	5.00/5.00	2017

PUBLICATIONS

Research Articles:

- Biswas, S., Ahmed, M., Das, P., **Niharika, I. T**., Das, S. C., Shayla, Mst. U., Mita, M. A., Biswas, S., Hasan, Md. R., Zaman, S., & Saleh, Md. A. (2025). Biofilms: Mechanisms, quorum sensing, antibiotic resistance, and implications in the food industry. Biologia, 80(7), 1803–1821.

  https://doi.org/10.1007/s11756-025-01953-0
- Mohammad Al-baruni Chowdhury, Md. Motashim Billah, Sara Mostafa, Sumaya Noor, **Israt Tamanna Niharika**, Md. Hadisur Rahman. Structural and Functional Impact of PTEN non-synonymous SNPs in autism spectrum disorder: A Computational Approach. [In Preparation]

RESEARCH & PROJECT

Research Intern- Molecular Biology and Protein Science Laboratory

University of Rajshahi | Supervisor: Professor Md. Abu Reza, PhD (NUS, Singapore)

- Conducted DNA extraction, gel electrophoresis, Drug Identification (Yaba, Heroin, Methamphetamine, and other dangerous drugs)
- Bacterial cell culture, Competent Cell Preparation, Transcription, chromatographic assays (HPLC)

B.SC Final Year Project Work

University of Rajshahi | Supervisor: Kazi Faisal Hoque, PhD (Ru, Bangladesh)

- Contributed to the project "In Silico Characterization of a PET-Degrading α/β Hydrolase from Comamonas testosterone, and Recombinant Vector Construction for Heterologous Expression in E. coli BL21(DE3)."

Bioinformatics & Computational Biology Research Experience

- Contributed to the project "Structural and Functional Impact of PTEN non-synonymous SNPs in autism spectrum disorder: A Computational Approach."
- Protein Design, Molecular Docking, SNP distributions, and clinical significance using Bioinformatics pipeline (NCBI)

Conferences & Workshops

- GEB journal club seminar on "The COVID Experience of an ID Physician from India". Held on 28th April 2024 at the University of Rajshahi, Bangladesh
- International Scientific Conference-2024 held on 27 April at the University of Rajshahi, Bangladesh. The theme was "International Conference on Advances and Challenges through Translational Research in Biological Sciences
- GEB journal club seminar on "How Computers are Revolutionizing Biology". Held on 14th March 2024 at the University of Rajshahi, Bangladesh"
- 3-day hands-on training on "Research Methodology and Computer-Aided Drug Design"(https://www.researchgate.net/lab/ABCD-Laboratory-Bangladesh-Md-Junaid)

Courses & Certifications

- Scienspur-Cell biology Fall 2024 Lakshmi Mittal & Family South Asian Institute at HARVARD UNIVERSITY
- Scienspur-Molecular Genetics Fall 2023 Lakshmi Mittal & Family South Asian Institute at HARVARD UNIVERSITY

Technical Competences

Microbiology, Molecular Biology & Wet-Lab Techniques:

- Sample preparation, Bacterial Cell Culture, Bacterial cell identification
- DNA extraction, Plasmid Extraction, Gel Electrophoresis, PCR, Sanger sequencing
- Restriction Digestion and ligation of Plasmid DNA, Competent Cell Preparation, Bacterial Transformation, SDS-PAGE, HPLC
- Aseptic handling and sterilization
- UV-Visible Spectroscopy, Colorimetric Test

Bioinformatics & Computational Biology

- PCR Primer Design, Codon Optimization
- Protein structure prediction - [AlphaFold v3, Homology Modeling], Molecular Docking, Drug Design
- Data analysis and visualization using: Microsoft Excel, Python, GraphPad Prism

Regulatory Awareness

- Knowledge of Good Laboratory Practices (GLP)
- Familiar with Good Manufacturing Practices (GMP) workflows
- Awareness of data integrity and compliance documentation standards

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Volunteer Experience & Community Engagement

- **Volunteer**, Bangladesh Biology Olympiad (2021–2023)
  Assisted in organizing academic events, coordinated with participants, and supported science outreach initiatives for students
- **Blood Donor**
  Regularly participated in voluntary blood donation programs, contributing to emergency healthcare support and community well-being
- **Member/Volunteer**, Nabojagoron Foundation (Educational Support Initiative)
  Engaged in activities focused on providing education and joyful learning experiences for underprivileged children, including mentoring and organizing learning sessions

References

Dr. Md Abu Reza (Principal Scientist)

- Molecular Biology and Protein Science Laboratory (MBPSL)
- Professor at University of Rajshahi
- rezaru@gmail.com

Dr. Kazi Faisal Hoque

- Professor at University of Rajshahi
- faisal.geb@ru.ac.bd